

## Qualification Pack



# Plumbing and Heating (WorldSkills)

QP Code: PSC/Q0123

Version: 1.0

NSQF Level: 4.5

Water Management and Plumbing Skill Council || Unit No-606, DLF Prime Tower  
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## Qualification Pack

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## Qualification Pack

### PSC/Q0123: Plumbing and Heating (WorldSkills)

#### Brief Job Description

A candidate who certified under the qualification Plumbing and Heating (WorldSkills) is primarily responsible for installation, repair and maintenance of plumbing and heating systems in residential, commercial and industrial settings. The technician will play a crucial role in ensuring the proper functioning of water supply, drainage, gas, heating, dry wall systems, pump operations etc. and contributing to the overall comfort and safety of occupants.

#### Personal Attributes

The individual must possess technical proficiency, mechanical aptitude, problem solving skills, thoroughness and precision in measurement and levelling. The person should be good at following instructions and a team player with a positive attitude.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [PSC/N0178: Workplace Management and Handling of Tools, Equipment and Materials](#)
2. [PSC/N0179: Analysis of Test Project](#)
3. [PSC/N0180: Conduct Operations on Pipes, Fittings and other sanitary fixtures for installation](#)
4. [PSC/N0181: Install Sanitary and CP fittings on Normal and Drywall System](#)
5. [PSC/N0182: Understand Marking and Assessment Criteria and Finishing and Handover the Job](#)
6. [PSC/N0136: Apply Health and Safety Practices at the Workplace](#)

#### Qualification Pack (QP) Parameters

|                   |                                                                           |
|-------------------|---------------------------------------------------------------------------|
| <b>Sector</b>     | Plumbing                                                                  |
| <b>Sub-Sector</b> | Industrial / Non-Industrial Plumbing, Gas & Piping (Industrial / Medical) |
| <b>Occupation</b> | Plumbing Systems Installation and Maintenance, Plumbing Servicing         |
| <b>Country</b>    | India                                                                     |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NSQF Level</b>                                         | 4.5                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Credits</b>                                            | 17                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/7126.0101                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Minimum Educational Qualification &amp; Experience</b> | <p>Pursuing 2nd year of UG (B.Tech / B.E. in Civil / Mechanical Engineering)</p> <p>OR</p> <p>Completed 2nd year diploma after 12th (Diploma in Civil / Mechanical Engineering )</p> <p>OR</p> <p>Completed 3 year diploma after 10th (Diploma in Civil or Mechanical Engineering)</p> <p>OR</p> <p>10th Class + I.T.I (Preferably in Plumber / Fitter Trade)</p> <p>OR</p> <p>12th Class/I.T.I (Preferably in Plumber / Fitter Trade)</p> |
| <b>Minimum Level of Education for Training in School</b>  | 10th Class                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Next Review Date</b>                                   | 31/01/2026                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>NSQC Approval Date</b>                                 | 31/01/2024                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Reference code on NQR</b>                              | QG-4.5-PL-01784-2024-V1-WMPS                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>NQR Version</b>                                        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Remarks:

The candidate must be aligned to specific age criteria as specified by WorldSkills for the trade Plumbing and Heating for the particular year of competition.

## Qualification Pack

# PSC/N0178: Workplace Management and Handling of Tools, Equipment and Materials

## Description

This NOS deals with systematic organization, coordination and optimization of resources, processes and people within the work environment to achieve specific goals and objectives as per the WorldSkills standards.

## Scope

The scope covers the following :

- Understand the workspace design and infrastructure
- Optimize workflow and process
- Identify tools, equipment and calculate quantity of materials as per drawing
- Evaluate and monitor the job assigned

## Elements and Performance Criteria

### *Understand the workspace design and infrastructure*

To be competent, the user/individual on the job must be able to:

- PC1.** manage workstation area as provided
- PC2.** implement solutions to enhance productivity
- PC3.** maintain tidiness of work station assigned
- PC4.** place tools, equipment and materials in proper stack as per the module assigned for the project

### *Optimize workflow and process*

To be competent, the user/individual on the job must be able to:

- PC5.** streamline work processes for efficiency
- PC6.** identify and eliminate discrepancies, if any, with proper approval from team management and juries
- PC7.** implement best practices for workflow management
- PC8.** keep tracking of timings as per the module assigned and complete the same within scheduled time

### *Identify tools, equipment and calculate quantity of materials as per drawing*

To be competent, the user/individual on the job must be able to:

- PC9.** review the drawing to identify specific tasks and operations
- PC10.** break down the materials into categories such as water supply, drainage, gas heating, towel rails, sanitary and CP fittings etc.
- PC11.** utilize measurements and dimensions provided in the drawing to calculate material quantities
- PC12.** create a comprehensive list that includes the name, quantity, specifications and any other relevant details for each tool, equipment and material

## Qualification Pack

### *Evaluate and monitor the job assigned*

To be competent, the user/individual on the job must be able to:

- PC13.** define the objectives and expectations of the assigned job precisely
- PC14.** ensure that the individual or team understands the goals they are expected to achieve
- PC15.** establish deadlines for key deliverables to track progress effectively
- PC16.** verify that the work aligns with the overall strategy

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** knowledge of work station area
- KU2.** understand processes and workflows
- KU3.** knowledge of process optimization techniques
- KU4.** understand of workplace safety standards and compliance
- KU5.** understand time saving techniques as per standards
- KU6.** understand the importance of accurate and detailed documentation
- KU7.** understand computer-aided design (CAD) symbols and units

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ability to articulate ideas clearly and effectively
- GS2.** problem solving skills
- GS3.** time management
- GS4.** adaptability towards changing circumstances and handle unexpected challenges
- GS5.** ability to collaborate with others, contribute to team goals and work cohesively
- GS6.** critical thinking ability
- GS7.** adherence to ethical standards and principles in all workplace activities
- GS8.** quality assurance
- GS9.** technical competence
- GS10.** risk managements
- GS11.** environmental sustainability



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Understand the workspace design and infrastructure</i>                                                                                     | <b>10</b>    | <b>15</b>       | -             | -          |
| <b>PC1.</b> manage workstation area as provided                                                                                               | -            | -               | -             | -          |
| <b>PC2.</b> implement solutions to enhance productivity                                                                                       | -            | -               | -             | -          |
| <b>PC3.</b> maintain tidiness of work station assigned                                                                                        | -            | -               | -             | -          |
| <b>PC4.</b> place tools, equipment and materials in proper stack as per the module assigned for the project                                   | -            | -               | -             | -          |
| <i>Optimize workflow and process</i>                                                                                                          | <b>10</b>    | <b>15</b>       | -             | -          |
| <b>PC5.</b> streamline work processes for efficiency                                                                                          | -            | -               | -             | -          |
| <b>PC6.</b> identify and eliminate discrepancies, if any, with proper approval from team management and juries                                | -            | -               | -             | -          |
| <b>PC7.</b> implement best practices for workflow management                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> keep tracking of timings as per the module assigned and complete the same within scheduled time                                   | -            | -               | -             | -          |
| <i>Identify tools, equipment and calculate quantity of materials as per drawing</i>                                                           | <b>10</b>    | <b>15</b>       | -             | -          |
| <b>PC9.</b> review the drawing to identify specific tasks and operations                                                                      | -            | -               | -             | -          |
| <b>PC10.</b> break down the materials into categories such as water supply, drainage, gas heating, towel rails, sanitary and CP fittings etc. | -            | -               | -             | -          |
| <b>PC11.</b> utilize measurements and dimensions provided in the drawing to calculate material quantities                                     | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> create a comprehensive list that includes the name, quantity, specifications and any other relevant details for each tool, equipment and material | -            | -               | -             | -          |
| <i>Evaluate and monitor the job assigned</i>                                                                                                                   | <b>10</b>    | <b>15</b>       | -             | -          |
| <b>PC13.</b> define the objectives and expectations of the assigned job precisely                                                                              | -            | -               | -             | -          |
| <b>PC14.</b> ensure that the individual or team understands the goals they are expected to achieve                                                             | -            | -               | -             | -          |
| <b>PC15.</b> establish deadlines for key deliverables to track progress effectively                                                                            | -            | -               | -             | -          |
| <b>PC16.</b> verify that the work aligns with the overall strategy                                                                                             | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                               | <b>40</b>    | <b>60</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| <b>NOS Code</b>            | PSC/N0178                                                           |
| <b>NOS Name</b>            | Workplace Management and Handling of Tools, Equipment and Materials |
| <b>Sector</b>              | Plumbing                                                            |
| <b>Sub-Sector</b>          | Industrial / Non-Industrial Plumbing                                |
| <b>Occupation</b>          | Plumbing Systems Installation and Maintenance                       |
| <b>NSQF Level</b>          | 4.5                                                                 |
| <b>Credits</b>             | 3                                                                   |
| <b>Version</b>             | 1.0                                                                 |
| <b>Last Reviewed Date</b>  | 31/01/2024                                                          |
| <b>Next Review Date</b>    | 31/01/2026                                                          |
| <b>NSQC Clearance Date</b> | 31/01/2024                                                          |

## Qualification Pack

### PSC/N0179: Analysis of Test Project

#### Description

This NOS deals with analysis of a plumbing drawing / test project which is a crucial step in understanding the layout and specifications for plumbing systems in a building or construction project.

#### Scope

The scope covers the following :

- Understand Symbols and Labels
- Identify Pipe Materials and Fittings
- Identify Parts of Sanitary and CP Fittings to Fix on Wall

#### Elements and Performance Criteria

##### *Understand Symbols and Labels*

To be competent, the user/individual on the job must be able to:

- PC1.** interpret plumbing symbols used in the drawing
- PC2.** identify different components such as pipes, fixtures, valves and other plumbing elements
- PC3.** recognize labels indicating materials, sizes and specification

##### *Identify Pipe Materials and Fittings*

To be competent, the user/individual on the job must be able to:

- PC4.** determine types of pipes (e.g. water supply, drainage, gas heating and towel rail)
- PC5.** identify pipe sizes, materials as per the project assigned
- PC6.** examine arrows or indicators showing the flow direction of water or wastewater
- PC7.** identify connection points between pipes and fixtures

##### *Identify Parts of Sanitary and CP Fittings to Fix on Wall*

To be competent, the user/individual on the job must be able to:

- PC8.** analyze the locations of plumbing fixtures (sinks, toilets, showers, etc.)
- PC9.** understand the connections and pathways between fixtures and the main plumbing system as per drawing
- PC10.** differentiate between hot and cold water pipes and connect to fixtures (wherever needed) along with valves and pumps as mentioned on drawing

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understand the specifications for different materials used in the plumbing system (e.g., copper, PVC, PEX etc.)
- KU2.** interpret isometric views to visualize the three-dimensional layout of the plumbing system
- KU3.** testing tools and technologies

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- KU4.** ability to identify potential risks associated with the test project and develop strategies for risk mitigation
- KU5.** knowledge of creating comprehensive test plans and strategies to ensure thorough test coverage
- KU6.** ability to track and manage defects efficiently along with generating comprehensive defect reports
- KU7.** knowledge of work station area
- KU8.** understand processes and workflows
- KU9.** knowledge of process optimization techniques
- KU10.** understand of workplace safety standards and compliance
- KU11.** understand time saving techniques as per standards
- KU12.** understand the importance of accurate and detailed documentation
- KU13.** understand computer-aided design (CAD) symbols and units

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ability to articulate ideas clearly and effectively
- GS2.** problem solving skills
- GS3.** time management
- GS4.** adaptability towards changing circumstances and handle unexpected challenges
- GS5.** ability to collaborate with others, contribute to team goals and work cohesively
- GS6.** critical thinking ability
- GS7.** adherence to ethical standards and principles in all workplace activities
- GS8.** quality assurance
- GS9.** technical competence
- GS10.** risk managements
- GS11.** environmental sustainability

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Understand Symbols and Labels</i>                                                                                                                      | <b>10</b>    | <b>20</b>       | -             | -          |
| <b>PC1.</b> interpret plumbing symbols used in the drawing                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify different components such as pipes, fixtures, valves and other plumbing elements                                                     | -            | -               | -             | -          |
| <b>PC3.</b> recognize labels indicating materials, sizes and specification                                                                                | -            | -               | -             | -          |
| <i>Identify Pipe Materials and Fittings</i>                                                                                                               | <b>10</b>    | <b>25</b>       | -             | -          |
| <b>PC4.</b> determine types of pipes (e.g. water supply, drainage, gas heating and towel rail)                                                            | -            | -               | -             | -          |
| <b>PC5.</b> identify pipe sizes, materials as per the project assigned                                                                                    | -            | -               | -             | -          |
| <b>PC6.</b> examine arrows or indicators showing the flow direction of water or wastewater                                                                | -            | -               | -             | -          |
| <b>PC7.</b> identify connection points between pipes and fixtures                                                                                         | -            | -               | -             | -          |
| <i>Identify Parts of Sanitary and CP Fittings to Fix on Wall</i>                                                                                          | <b>10</b>    | <b>25</b>       | -             | -          |
| <b>PC8.</b> analyze the locations of plumbing fixtures (sinks, toilets, showers, etc.)                                                                    | -            | -               | -             | -          |
| <b>PC9.</b> understand the connections and pathways between fixtures and the main plumbing system as per drawing                                          | -            | -               | -             | -          |
| <b>PC10.</b> differentiate between hot and cold water pipes and connect to fixtures (wherever needed) along with valves and pumps as mentioned on drawing | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                          | <b>30</b>    | <b>70</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                               |
|----------------------------|-----------------------------------------------|
| <b>NOS Code</b>            | PSC/N0179                                     |
| <b>NOS Name</b>            | Analysis of Test Project                      |
| <b>Sector</b>              | Plumbing                                      |
| <b>Sub-Sector</b>          | Industrial / Non-Industrial Plumbing          |
| <b>Occupation</b>          | Plumbing Systems Installation and Maintenance |
| <b>NSQF Level</b>          | 4.5                                           |
| <b>Credits</b>             | 3                                             |
| <b>Version</b>             | 1.0                                           |
| <b>Last Reviewed Date</b>  | 31/01/2024                                    |
| <b>Next Review Date</b>    | 31/01/2026                                    |
| <b>NSQC Clearance Date</b> | 31/01/2024                                    |

## Qualification Pack

### PSC/N0180: Conduct Operations on Pipes, Fittings and other sanitary fixtures for installation

#### Description

This NOS deals with the installation of water supply pipes, drainage pipes, gas heating and towel rails which involves a systematic and precise process to ensure the efficient and safe conveyance of water

#### Scope

The scope covers the following :

- Selection of materials as per the drawing and standards
- Preparation for piping process
- Conduct operations on pipes
- Install Pipes with its fittings
- Install Valves and Pumps

#### Elements and Performance Criteria

##### *Selection of materials as per the drawing and standards*

To be competent, the user/individual on the job must be able to:

- PC1.** examine the drawings and specifications provided
- PC2.** identify the functional requirements of the part or product
- PC3.** identify tools and materials for different modules as per the test project

##### *Preparation for piping process*

To be competent, the user/individual on the job must be able to:

- PC4.** determine the fitting requirements to install the pipes
- PC5.** measure and mark dimensions for fabrication on the pipes and fittings making allowances for spring-back, distortion and assembly

##### *Conduct operations on pipes Install Pipes with its fittings*

To be competent, the user/individual on the job must be able to:

- PC6.** cut various types of pipes to the length determined with proper tools
- PC7.** cut threads on GI pipe ends to the appropriate length as per dimension and method of assessment
- PC8.** bend and form the pipes to specified angle and offsets as mentioned in drawing
- PC9.** prepare pipe ends using techniques such as reaming, leveling, filing and grinding
- PC10.** join different types of pipes using fittings and relevant techniques
- PC11.** ensure the assembled pipe meets the required specifications
- PC12.** report any difficulties that may arise and carry out remedial actions
- PC13.** identify the area and route where the assembled pipes are to be installed on board
- PC14.** install water supply pipes, drainage pipes, gas heating pipes and towel rail pipes and other water supply components according to given specifications and site requirements without damage or interference to the surrounding structures

## Qualification Pack

**PC15.** check if all the water supply system installations are properly aligned, sized, supported and functioning properly

### *Install Valves and Pumps*

To be competent, the user/individual on the job must be able to:

**PC16.** identify different types of valves as per the drawing

**PC17.** identify pumps to be installed as per the project assigned

**PC18.** install ball valves and Thermostatic Mixing Valve (TMV) as per standards

**PC19.** install recirculation pumps as per standards

**PC20.** check the valves and pumps installed whether in proper position and dimensions mentioned on the drawing

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** understand marking and assessment criteria prepared as per worldskills and indiaskills standards

**KU2.** knowledge of work station area

**KU3.** understand processes and workflows

**KU4.** knowledge of process optimization techniques

**KU5.** understand of workplace safety standards and compliance

**KU6.** understand time saving techniques as per standards

**KU7.** understand the importance of accurate and detailed documentation

**KU8.** understand computer-aided design (CAD) symbols and units

**KU9.** understand usages of tools and techniques

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** ability to articulate ideas clearly and effectively

**GS2.** problem solving skills

**GS3.** time management

**GS4.** adaptability towards changing circumstances and handle unexpected challenges

**GS5.** ability to collaborate with others, contribute to team goals and work cohesively

**GS6.** critical thinking ability

**GS7.** adherence to ethical standards and principles in all workplace activities

**GS8.** quality assurance

**GS9.** technical competence

**GS10.** risk managements

**GS11.** environmental sustainability



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Selection of materials as per the drawing and standards</i>                                                                               | 5            | 10              | -             | -          |
| <b>PC1.</b> examine the drawings and specifications provided                                                                                 | -            | -               | -             | -          |
| <b>PC2.</b> identify the functional requirements of the part or product                                                                      | -            | -               | -             | -          |
| <b>PC3.</b> identify tools and materials for different modules as per the test project                                                       | -            | -               | -             | -          |
| <i>Preparation for piping process</i>                                                                                                        | 5            | 10              | -             | -          |
| <b>PC4.</b> determine the fitting requirements to install the pipes                                                                          | -            | -               | -             | -          |
| <b>PC5.</b> measure and mark dimensions for fabrication on the pipes and fittings making allowances for spring-back, distortion and assembly | -            | -               | -             | -          |
| <i>Conduct operations on pipes Install Pipes with its fittings</i>                                                                           | 10           | 25              | -             | -          |
| <b>PC6.</b> cut various types of pipes to the length determined with proper tools                                                            | -            | -               | -             | -          |
| <b>PC7.</b> cut threads on GI pipe ends to the appropriate length as per dimension and method of assessment                                  | -            | -               | -             | -          |
| <b>PC8.</b> bend and form the pipes to specified angle and offsets as mentioned in drawing                                                   | -            | -               | -             | -          |
| <b>PC9.</b> prepare pipe ends using techniques such as reaming, leveling, filing and grinding                                                | -            | -               | -             | -          |
| <b>PC10.</b> join different types of pipes using fittings and relevant techniques                                                            | -            | -               | -             | -          |
| <b>PC11.</b> ensure the assembled pipe meets the required specifications                                                                     | -            | -               | -             | -          |
| <b>PC12.</b> report any difficulties that may arise and carry out remedial actions                                                           | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> identify the area and route where the assembled pipes are to be installed on board                                                                                                                                                        | -            | -               | -             | -          |
| <b>PC14.</b> install water supply pipes, drainage pipes, gas heating pipes and towel rail pipes and other water supply components according to given specifications and site requirements without damage or interference to the surrounding structures | -            | -               | -             | -          |
| <b>PC15.</b> check if all the water supply system installations are properly aligned, sized, supported and functioning properly                                                                                                                        | -            | -               | -             | -          |
| <i>Install Valves and Pumps</i>                                                                                                                                                                                                                        | <b>10</b>    | <b>25</b>       | -             | -          |
| <b>PC16.</b> identify different types of valves as per the drawing                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC17.</b> identify pumps to be installed as per the project assigned                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC18.</b> install ball valves and Thermostatic Mixing Valve (TMV) as per standards                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC19.</b> install recirculation pumps as per standards                                                                                                                                                                                              | -            | -               | -             | -          |
| <b>PC20.</b> check the valves and pumps installed whether in proper position and dimensions mentioned on the drawing                                                                                                                                   | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                       | <b>30</b>    | <b>70</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                    |
|----------------------------|------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | PSC/N0180                                                                          |
| <b>NOS Name</b>            | Conduct Operations on Pipes, Fittings and other sanitary fixtures for installation |
| <b>Sector</b>              | Plumbing                                                                           |
| <b>Sub-Sector</b>          | Industrial / Non-Industrial Plumbing                                               |
| <b>Occupation</b>          | Plumbing Systems Installation and Maintenance                                      |
| <b>NSQF Level</b>          | 4.5                                                                                |
| <b>Credits</b>             | 6                                                                                  |
| <b>Version</b>             | 1.0                                                                                |
| <b>Last Reviewed Date</b>  | 31/01/2024                                                                         |
| <b>Next Review Date</b>    | 31/01/2026                                                                         |
| <b>NSQC Clearance Date</b> | 31/01/2024                                                                         |

## Qualification Pack

### PSC/N0181: Install Sanitary and CP fittings on Normal and Drywall System

#### Description

This NOS deals in detail with installation of plumbing fixtures in accordance with industry standards and project requirements on both normal and dry wall system.

#### Scope

The scope covers the following :

- Prepare for installation of fixtures
- Install sanitary fixtures, support and related accessories
- Install drywall system

#### Elements and Performance Criteria

##### *Prepare for installation of fixtures*

To be competent, the user/individual on the job must be able to:

- PC1.** tally the count and quality of fixtures, parts, support material provided in the packing with the manufacturer's list or order form
- PC2.** select size, type and quantity of fixture and trim for specific application according to given specifications
- PC3.** locate and mark the position of fixtures and fixture supports in structures

##### *Install sanitary fixtures, support and related accessories*

To be competent, the user/individual on the job must be able to:

- PC4.** cut the physical structures such as wooden walls with correct markings and specifications for performing pipe installations
- PC5.** install supports for plumbing and heating piping using appropriate fasteners to ensure grade, spacing and alignment
- PC6.** install fixtures as per specifications without damage to fixture, pipework, the surrounding environment etc
- PC7.** Install and secure the sensor for touchless fittings and fixtures in the slot provided slot (in case added in the test project only)
- PC8.** plumb and level the supports and fixtures installed to ensure alignment and balance
- PC9.** identify and rectify faults that are within limits of expertise and authority and escalate others to authorized personnel
- PC10.** clear the work area and dispose, reuse or recycle left over materials according to regulations and codes of practice

##### *Install drywall system*

To be competent, the user/individual on the job must be able to:

- PC11.** perform measurements for cutting of dry wall channels
- PC12.** drive screws into drywall panels
- PC13.** remove any excess drywall paneling

## Qualification Pack

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** knowledge of work station area
- KU2.** understand processes and workflows
- KU3.** knowledge of process optimization techniques
- KU4.** understand of workplace safety standards and compliance
- KU5.** understand time saving techniques as per standards
- KU6.** understand the importance of accurate and detailed documentation
- KU7.** understand computer-aided design (CAD) symbols and units
- KU8.** alignment and elevation techniques used in plumbing and heating systems
- KU9.** alignment and elevation techniques used in dry wall systems
- KU10.** types of WCs, Wash basins, kitchen sinks and CP fittings and its description
- KU11.** correct practices for installing plumbing fixture
- KU12.** codes, standards and regulations applicable for the installation of plumbing fixtures

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** measure all dimensions used in plumbing in metric scale
- GS2.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS3.** ability to articulate ideas clearly and effectively
- GS4.** problem solving skills
- GS5.** time management
- GS6.** adaptability towards changing circumstances and handle unexpected challenges
- GS7.** ability to collaborate with others, contribute to team goals and work cohesively
- GS8.** critical thinking ability
- GS9.** adherence to ethical standards and principles in all workplace activities
- GS10.** quality assurance
- GS11.** technical competence
- GS12.** risk managements
- GS13.** environmental sustainability

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare for installation of fixtures</i>                                                                                                      | <b>10</b>    | <b>15</b>       | -             | -          |
| <b>PC1.</b> tally the count and quality of fixtures, parts, support material provided in the packing with the manufacturer's list or order form  | -            | -               | -             | -          |
| <b>PC2.</b> select size, type and quantity of fixture and trim for specific application according to given specifications                        | -            | -               | -             | -          |
| <b>PC3.</b> locate and mark the position of fixtures and fixture supports in structures                                                          | -            | -               | -             | -          |
| <i>Install sanitary fixtures, support and related accessories</i>                                                                                | <b>10</b>    | <b>35</b>       | -             | -          |
| <b>PC4.</b> cut the physical structures such as wooden walls with correct markings and specifications for performing pipe installations          | -            | -               | -             | -          |
| <b>PC5.</b> install supports for plumbing and heating piping using appropriate fasteners to ensure grade, spacing and alignment                  | -            | -               | -             | -          |
| <b>PC6.</b> install fixtures as per specifications without damage to fixture, pipework, the surrounding environment etc                          | -            | -               | -             | -          |
| <b>PC7.</b> Install and secure the sensor for touchless fittings and fixtures in the slot provided slot (in case added in the test project only) | -            | -               | -             | -          |
| <b>PC8.</b> plumb and level the supports and fixtures installed to ensure alignment and balance                                                  | -            | -               | -             | -          |
| <b>PC9.</b> identify and rectify faults that are within limits of expertise and authority and escalate others to authorized personnel            | -            | -               | -             | -          |
| <b>PC10.</b> clear the work area and dispose, reuse or recycle left over materials according to regulations and codes of practice                | -            | -               | -             | -          |
| <i>Install drywall system</i>                                                                                                                    | <b>10</b>    | <b>20</b>       | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> perform measurements for cutting of dry wall channels | -            | -               | -             | -          |
| <b>PC12.</b> drive screws into drywall panels                      | -            | -               | -             | -          |
| <b>PC13.</b> remove any excess drywall paneling                    | -            | -               | -             | -          |
| <b>NOS Total</b>                                                   | <b>30</b>    | <b>70</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                               |
|----------------------------|---------------------------------------------------------------|
| <b>NOS Code</b>            | PSC/N0181                                                     |
| <b>NOS Name</b>            | Install Sanitary and CP fittings on Normal and Drywall System |
| <b>Sector</b>              | Plumbing                                                      |
| <b>Sub-Sector</b>          | Industrial / Non-Industrial Plumbing                          |
| <b>Occupation</b>          | Plumbing Systems Installation and Maintenance                 |
| <b>NSQF Level</b>          | 4.5                                                           |
| <b>Credits</b>             | 4                                                             |
| <b>Version</b>             | 1.0                                                           |
| <b>Last Reviewed Date</b>  | 31/01/2024                                                    |
| <b>Next Review Date</b>    | 31/01/2026                                                    |
| <b>NSQC Clearance Date</b> | 31/01/2024                                                    |



## Qualification Pack

# PSC/N0182: Understand Marking and Assessment Criteria and Finishing and Handover the Job

## Description

This NOS deals with understanding of marking and assessment criteria designed by the council and handing over the job to juries at the end of the competition

## Scope

The scope covers the following :

- Understand marking and assessment criteria set by council
- Learn the handing over process

## Elements and Performance Criteria

### *Understand marking and assessment criteria set by council*

To be competent, the user/individual on the job must be able to:

- PC1.** understand the module wise marking and assessment criteria
- PC2.** understand the assignment of marks as per the plumb level and gradient
- PC3.** understand the assignment of marks as per the dimension
- PC4.** understand the assignment of marks as per the angles
- PC5.** handle the tools and equipment properly

### *Learn the handing over process*

To be competent, the user/individual on the job must be able to:

- PC6.** complete each module of the project in the assigned time
- PC7.** handover the project assigned post to the completion of job
- PC8.** respond to questionnaires of juries

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** escalation matrix and procedures for reporting work
- KU2.** types of harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it
- KU3.** importance of effective communication and the impact of poor communication on the candidate
- KU4.** various components of effective communication such as tone and pitch
- KU5.** how to express and address grievances appropriately and effectively
- KU6.** laws, acts and provisions defined for PwD by the statutory bodies
- KU7.** gender, disability and cultural biases, stereotypes and impact on others
- KU8.** importance of teamwork in organizational and individual success

## Qualification Pack

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes, keep records, prepare to-do lists and take down instructions
- GS2.** write basic numbers, quantities and basic work-related terminology for operational requirements in the local language and English
- GS3.** read basic terminologies to accurately interpret labels, supervisor's instructions in the local language and English
- GS4.** read and interpret accurate information from work-related documents and various relevant work instructions and records in local
- GS5.** interact with the concerned personnel appropriately (correct protocol and manner of speaking etc.)
- GS6.** display active listening skills while interacting with co-workers and others in the workplace
- GS7.** deliver consistent and reliable service to internal and external customers
- GS8.** work with co-workers and supervisor to resolve any issues that threaten work quality as per the planned schedule

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Understand marking and assessment criteria set by council</i>                   | 10           | 15              | -             | -          |
| <b>PC1.</b> understand the module wise marking and assessment criteria             | -            | -               | -             | -          |
| <b>PC2.</b> understand the assignment of marks as per the plumb level and gradient | -            | -               | -             | -          |
| <b>PC3.</b> understand the assignment of marks as per the dimension                | -            | -               | -             | -          |
| <b>PC4.</b> understand the assignment of marks as per the angles                   | -            | -               | -             | -          |
| <b>PC5.</b> handle the tools and equipment properly                                | -            | -               | -             | -          |
| <i>Learn the handing over process</i>                                              | 10           | 15              | -             | -          |
| <b>PC6.</b> complete each module of the project in the assigned time               | -            | -               | -             | -          |
| <b>PC7.</b> handover the project assigned post to the completion of job            | -            | -               | -             | -          |
| <b>PC8.</b> respond to questionnaires of juries                                    | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                   | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                               |
|----------------------------|-------------------------------------------------------------------------------|
| <b>NOS Code</b>            | PSC/N0182                                                                     |
| <b>NOS Name</b>            | Understand Marking and Assessment Criteria and Finishing and Handover the Job |
| <b>Sector</b>              | Plumbing                                                                      |
| <b>Sub-Sector</b>          | Industrial / Non-Industrial Plumbing                                          |
| <b>Occupation</b>          | Plumbing Systems Installation and Maintenance                                 |
| <b>NSQF Level</b>          | 4.5                                                                           |
| <b>Credits</b>             | 0.5                                                                           |
| <b>Version</b>             | 1.0                                                                           |
| <b>Last Reviewed Date</b>  | 31/01/2024                                                                    |
| <b>Next Review Date</b>    | 31/01/2026                                                                    |
| <b>NSQC Clearance Date</b> | 31/01/2024                                                                    |

## Qualification Pack

### PSC/N0136: Apply Health and Safety Practices at the Workplace

#### Description

This unit is about practices to be executed at the workplace which is crucial for creating a safe and secure working environment that promotes the well-being of employees and minimizes the risk of accidents, injuries, and occupational illnesses. Here's a description outlining the key aspects of implementing health and safety practices at the workplace.

#### Scope

The scope covers the following :

- Follow safety measures to avoid accidents
- Daily inspection of tools and equipment
- Deal with emergencies

#### Elements and Performance Criteria

##### *Follow safety measures to avoid accidents*

To be competent, the user/individual on the job must be able to:

- PC1.** identify risks and hazards at the workplace
- PC2.** wear personal protective equipment (PPE) as per the type of plumbing work
- PC3.** place protective barricades and signages around the pits and trenches
- PC4.** isolate the plumbing fittings and fixtures from electrical wiring to avoid accidents
- PC5.** adhere to organisational procedures for reporting hazards and incidents to relevant authorities
- PC6.** establish ventilation before entering underground work areas
- PC7.** work safely in and around trenches, elevated places and confined areas
- PC8.** ensure tools and hazardous materials are not left unattended
- PC9.** ensure good housekeeping in order to prevent hazards e.g. fire
- PC10.** dispose waste materials and used PPE according to regulations and codes of practice

##### *Follow hygiene and sanitation practices*

To be competent, the user/individual on the job must be able to:

- PC11.** follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with a bent elbow while coughing/sneezing etc.
- PC12.** clean and disinfect work area, materials/supplies, equipment etc. before and after use.
- PC13.** report hygiene and sanitation issues to appropriate authority

##### *Use tools, equipment and materials safely*

To be competent, the user/individual on the job must be able to:

- PC14.** check that the tools, equipment and materials are in good condition and as per industry standards before use
- PC15.** use power tools and machinery that are grounded
- PC16.** replace or repair split or loose tools before use

## Qualification Pack

**PC17.** store and transport various plumbing materials safely

### *Deal with emergencies*

To be competent, the user/individual on the job must be able to:

**PC18.** follow workplace emergency and evacuation procedures

**PC19.** use a fire extinguisher correctly

**PC20.** use safe methods to free a person from electrocution

**PC21.** administer appropriate first aid (such as CPR etc.) to victims in case of cuts, bleeding, burns, choking, electric shock, poisoning etc.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** possible causes of risk, hazard or accident in the workplace

**KU2.** organisational procedures for upkeep of tools and plumbing materials, health and safety

**KU3.** location of all the general health and safety equipment in the workplace

**KU4.** meaning of hazards and risks

**KU5.** hazardous environment encountered during work such as underground areas, elevated areas, areas with water and electricity supply, presence of biological waste, under construction sites etc.

**KU6.** work practices and precautions to control and prevent risks, hazards and accidents

**KU7.** importance of each personal protective equipment used such as eye protection mask, hard hats, gloves, apron, rubber boots etc.

**KU8.** tools and plumbing equipment as per latest industry standards

**KU9.** preventative and remedial actions to be taken in case of exposure to toxic materials

**KU10.** specific safety and health related problems faced in domestic, commercial and institutional setups

**KU11.** various causes of fire and precautionary activities to prevent the fire accident

**KU12.** techniques of using the different fire extinguishers

**KU13.** rescue techniques applied during a fire hazard

**KU14.** various types of safety signs and meaning

**KU15.** appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, minor burns, poisoning, eye injuries etc.

**KU16.** potential injuries and ill health associated with incorrect handling of tools and equipment

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** write an accident/incident report in local language or English

**GS2.** read and comprehend labels, charts, signages, manuals, plumbing symbols etc.

**GS3.** question coworkers appropriately in order to clarify instructions and other issues



## Qualification Pack

- GS4.** plan and organize the work schedule, work area, tools, equipment and materials for improved productivity
- GS5.** determine key considerations and priorities when faced with problems
- GS6.** seek official and authorised sources of help and guidance to resolve problems that cannot be solved at one's level of authority

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Follow safety measures to avoid accidents</i>                                                                                                                               | <b>12</b>    | <b>32</b>       | -             | <b>5</b>   |
| <b>PC1.</b> identify risks and hazards at the workplace                                                                                                                        | 1            | 2               | -             | 0.5        |
| <b>PC2.</b> wear personal protective equipment (PPE) as per the type of plumbing work                                                                                          | 1            | 2               | -             | 0.5        |
| <b>PC3.</b> place protective barricades and signages around the pits and trenches                                                                                              | 1            | 4               | -             | 0.5        |
| <b>PC4.</b> isolate the plumbing fittings and fixtures from electrical wiring to avoid accidents                                                                               | 1            | 4               | -             | 0.5        |
| <b>PC5.</b> adhere to organisational procedures for reporting hazards and incidents to relevant authorities                                                                    | 1            | 4               | -             | 0.5        |
| <b>PC6.</b> establish ventilation before entering underground work areas                                                                                                       | 1            | 4               | -             | 0.5        |
| <b>PC7.</b> work safely in and around trenches, elevated places and confined areas                                                                                             | 1            | 3               | -             | 0.5        |
| <b>PC8.</b> ensure tools and hazardous materials are not left unattended                                                                                                       | 1            | 3               | -             | 0.5        |
| <b>PC9.</b> ensure good housekeeping in order to prevent hazards e.g. fire                                                                                                     | 2            | 3               | -             | 0.5        |
| <b>PC10.</b> dispose waste materials and used PPE according to regulations and codes of practice                                                                               | 2            | 3               | -             | 0.5        |
| <i>Follow hygiene and sanitation practices</i>                                                                                                                                 | <b>3</b>     | <b>8</b>        | -             | <b>2</b>   |
| <b>PC11.</b> follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with a bent elbow while coughing/sneezing etc. | 1            | 3               | -             | 1          |
| <b>PC12.</b> clean and disinfect work area, materials/supplies, equipment etc. before and after use.                                                                           | 1            | 3               | -             | 0.5        |
| <b>PC13.</b> report hygiene and sanitation issues to appropriate authority                                                                                                     | 1            | 2               | -             | 0.5        |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Use tools, equipment and materials safely</i>                                                                                                      | <b>7</b>     | <b>8</b>        | -             | <b>1</b>   |
| <b>PC14.</b> check that the tools, equipment and materials are in good condition and as per industry standards before use                             | 2            | 2               | -             | 0.25       |
| <b>PC15.</b> use power tools and machinery that are grounded                                                                                          | 2            | 2               | -             | 0.25       |
| <b>PC16.</b> replace or repair split or loose tools before use                                                                                        | 1            | 2               | -             | 0.25       |
| <b>PC17.</b> store and transport various plumbing materials safely                                                                                    | 2            | 2               | -             | 0.25       |
| <i>Deal with emergencies</i>                                                                                                                          | <b>8</b>     | <b>12</b>       | -             | <b>2</b>   |
| <b>PC18.</b> follow workplace emergency and evacuation procedures                                                                                     | 2            | 3               | -             | 0.5        |
| <b>PC19.</b> use a fire extinguisher correctly                                                                                                        | 2            | 3               | -             | 0.5        |
| <b>PC20.</b> use safe methods to free a person from electrocution                                                                                     | 2            | 3               | -             | 0.5        |
| <b>PC21.</b> administer appropriate first aid (such as CPR etc.) to victims in case of cuts, bleeding, burns, choking, electric shock, poisoning etc. | 2            | 3               | -             | 0.5        |
| <b>NOS Total</b>                                                                                                                                      | <b>30</b>    | <b>60</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                                                                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | PSC/N0136                                                                                                                                                                                                                                                                                                                   |
| <b>NOS Name</b>            | Apply Health and Safety Practices at the Workplace                                                                                                                                                                                                                                                                          |
| <b>Sector</b>              | Plumbing                                                                                                                                                                                                                                                                                                                    |
| <b>Sub-Sector</b>          | Industrial / Non-Industrial Plumbing, Water Supply & Water Treatment & Quality Control, Sewerage & Sewage Treatment, Drainage, Water Harvesting & Ground Recharging, Firefighting & Safety Systems, Gas & Piping (Industrial / Medical), HVAC & Steam, Manufacturing of Plumbing / Firefighting Products, Micro-Hydel Power |
| <b>Occupation</b>          | Plumbing Systems Installation and Maintenance, Designing, Sales, Material Handling, Plumbing Servicing, Plumbing Plant Operations                                                                                                                                                                                           |
| <b>NSQF Level</b>          | 4                                                                                                                                                                                                                                                                                                                           |
| <b>Credits</b>             | 0.5                                                                                                                                                                                                                                                                                                                         |
| <b>Version</b>             | 2.0                                                                                                                                                                                                                                                                                                                         |
| <b>Last Reviewed Date</b>  | 18/02/2025                                                                                                                                                                                                                                                                                                                  |
| <b>Next Review Date</b>    | 18/02/2028                                                                                                                                                                                                                                                                                                                  |
| <b>NSQC Clearance Date</b> | 18/02/2025                                                                                                                                                                                                                                                                                                                  |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each element will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Practical for each element.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at

## Qualification Pack

each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

### Minimum Aggregate Passing % at QP Level : 70

**(Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

Compulsory NOS

| National Occupational Standards                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|----------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| PSC/N0178.Workplace Management and Handling of Tools, Equipment and Materials                | 40           | 60              | 0             | 0          | 100         | 20         |
| PSC/N0179.Analysis of Test Project                                                           | 30           | 70              | 0             | 0          | 100         | 15         |
| PSC/N0180.Conduct Operations on Pipes, Fittings and other sanitary fixtures for installation | 30           | 70              | 0             | 0          | 100         | 30         |
| PSC/N0181.Install Sanitary and CP fittings on Normal and Drywall System                      | 30           | 70              | 0             | 0          | 100         | 25         |
| PSC/N0182.Understand Marking and Assessment Criteria and Finishing and Handover the Job      | 20           | 30              | 0             | 0          | 50          | 5          |
| PSC/N0136.Apply Health and Safety Practices at the Workplace                                 | 30           | 60              | -             | 10         | 100         | 5          |
| <b>Total</b>                                                                                 | <b>180</b>   | <b>360</b>      | <b>-</b>      | <b>10</b>  | <b>550</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |